CHAPTER 4

CLASSIFICATION OF DATA

CHAPTER ESSENCE AT A GLANCE:
Master data condensation: chronological, spatial, qualitative, and quantitative classification, frequency distributions, and class interval conventions.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Meaning and Objectives of Classification:** Classification arranges raw data into homogeneous groups based on shared attributes, condensing volume, facilitating comparison, and revealing distributions.

• **Four Modes of Classification:** Chronological (classified by time), Spatial (geographical area), Qualitative (attributes like gender, literacy), and Quantitative (numerical characteristics like income, marks).

• **Frequency Distribution Architecture:** Raw observations are tallied into discrete frequency distributions (individual values with counts) or continuous grouped distributions (class intervals with frequencies).

• **Class Conventions: Inclusive vs Exclusive:** Exclusive method (10-20, 20-30: upper limit excluded) maintains continuity; Inclusive method (10-19, 20-29: upper limit included) requires correction factor for continuity.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Class Mid-Value (Class Mark)} = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$$

Midpoint of Grouped Class Interval

$$\text{Class Width (Magnitude)} \iff h = \text{Upper Limit} - \text{Lower Limit}$$

Span of Continuous Interval Category

$$\text{Sturges Rule for Classes} \iff k = 1 + 3.322 \log_{10} N$$

Optimal Class Interval Count Estimation

$$\text{Inclusive Correction Factor} = \frac{\text{Lower Limit}_2 - \text{Upper Limit}_1}{2}$$

Adjustment Factor Converting Inclusive to Exclusive Limits

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In the Exclusive class method (e.g. 10-20), an observation of exactly 20 is counted in the 20-30 class, never in 10-20.

• **Alert:** In continuous frequency distributions, open-ended classes ('Below 20' or 'Above 80') make it impossible to compute Arithmetic Mean without arbitrary assumptions, though Median and Mode remain calculable.

• **Alert:** Always state class mark as the arithmetic mean of lower limit and upper limit.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is meant by classification of data? State any two primary objectives of classification. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The process of arranging collected raw data into homogeneous classes or groups according to their common characteristics or affinities. **[1 Mark]**

• **Two Objectives:** (1) To condense mass data into a manageable form; (2) To facilitate quick comparison between different groups and prepare data for tabulation. **[1 Mark]**

2. Explain the four different bases of classification of data with an example of each. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Chronological Classification:** Data arranged on the basis of time (years, months, days). Example: India's annual food grain production from 2015 to 2024. **[1 Mark]**

• **2. Spatial (Geographical) Classification:** Data arranged on the basis of location or geography. Example: State-wise literacy rates across Kerala, Tamil Nadu, Karnataka. **[1 Mark]**

• **3. Qualitative Classification:** Data grouped according to non-measurable qualitative attributes. Example: Classifying employees by gender (Male, Female) or educational attainment. **[1 Mark]**

• **4. Quantitative Classification:** Data arranged according to measurable numerical magnitudes. Example: Classifying students by marks obtained (0-10, 10-20, 20-30). **[1 Mark]**

3. Distinguish between the Exclusive Method and Inclusive Method of forming class intervals with examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Exclusive Method:** The upper limit of one class is the lower limit of the next class (e.g. 10-20, 20-30). An observation equal to the upper limit is excluded from that class and counted in the next. Maintains continuity. **[1.5 Marks]**

• **Inclusive Method:** The upper limit of a class is included within that class (e.g. 10-19, 20-29). Observation 19 is counted in 10-19. To convert to continuous boundaries, subtract 0.5 from lower limits and add 0.5 to upper limits (9.5-19.5, 19.5-29.5). **[1.5 Marks]**

4. Define: (a) Class Mark, (b) Class Frequency. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) Class Mark:** The midpoint of a class interval, computed as (Lower Class Limit + Upper Class Limit) divided by 2. **[1 Mark]**

• **(b) Class Frequency:** The total number of individual observations or tally counts that fall within a specified class interval. **[1 Mark]**

5. What is a Cumulative Frequency Distribution? Distinguish between 'Less than' and 'More than' types. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** A distribution formed by continuously adding successive class frequencies to express total observations above or below specified class boundaries. **[1 Mark]**

• **Less Than Cumulative Frequency:** Cumulates frequencies starting from the lowest class upward, indicating number of observations having values less than the upper limit of each class. **[1 Mark]**

• **More Than Cumulative Frequency:** Cumulates frequencies starting from the highest class downward, indicating number of observations having values more than or equal to the lower limit of each class. **[1 Mark]**

6. What are Open-Ended Classes? Name one statistical measure that cannot be computed for open-ended data without assumption. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Open-Ended Classes:** Classes where the lower limit of the first class is not specified (e.g. 'Below 20') or the upper limit of the last class is not specified (e.g. 'Above 80'). **[1 Mark]**

• **Measure Affected:** Arithmetic Mean cannot be computed because class marks for open-ended intervals cannot be determined without arbitrary assumptions. **[1 Mark]**